PAVAN KUMAR SIVALASETTY

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PROFESSIONAL SUMMARY

Seasoned AI & Machine Learning Engineer with 5+ years of hands-on experience designing, developing, and deploying scalable ML/AI systems in enterprise environments. Expert in Python, MLOps, and cloud platforms (AWS, GCP, Azure), with a specialized focus on LLM integration, prompt engineering, and vector search pipelines. Adept at building agentic AI systems, retrieval-augmented generation (RAG), and delivering reusable Python packages for ML lifecycle management. Proven record of delivering up to 60% increases in operational efficiency and collaborating cross-functionally to build robust, sustainable AI solutions for high-impact use cases.

PROFESSIONAL EXPERIENCE

AI/ML Engineer
CVS Pharmacy – Remote
March 2024 – Present

- Architected and deployed a GenAI-powered chatbot platform integrated with AWS Bedrock, OpenAI GPT-4 Turbo, and LangChain, handling 10K+ monthly user queries and automating document processing and compliance checks.
- Designed and implemented vectorized semantic search pipelines using Titan Text Embeddings, SageMaker, and FAISS, increasing contextual response accuracy by 65% across financial and legal use cases.
- Developed NLP microservices for intent classification, sentiment analysis, and entity recognition, reducing manual processing time by 40%.
- Built backend APIs using Django Rest Framework and FastAPI to manage LLM prompts, user sessions, and retrieval-based answering logic, ensuring seamless integration with web and mobile platforms.
- Designed multi-turn conversational memory storage with MongoDB, supporting personalized user experiences and increasing response satisfaction by 30%.
- Enabled scalable deployment leveraging Dockerized microservices, Kubernetes, and AWS EKS, maintaining 99.99% uptime during peak loads.
- Integrated CloudWatch, Prometheus, and Grafana for monitoring and real-time issue response, decreasing incident response time by 50%.
- Built continuous feedback and fine-tuning pipelines, ingesting over 500MB/day of data, and automated CI/CD with >85% test coverage, reducing release time by 70%.
- Collaborated with cross-functional teams in an Agile environment, contributing to sprint planning, story writing, and demos across 5+ major AI modules.

Outcome: Delivered a production-grade chatbot, improving operational efficiency by 60% and reducing compliance turnaround time by 45%.
TOOLS: Python, Django, FastAPI, MongoDB, AWS SageMaker, Bedrock, Lambda, EC2, S3, EKS, Hugging Face, LangChain, Docker, Kubernetes, GitHub Actions, Grafana, Prometheus, PyTest.

AI/ML Engineer
Guild Mortgage, San Diego, CA
April 2021 – December 2022

- Developed 15+ ML models using Vertex AI Pipelines with model accuracies exceeding 92%, reducing iteration cycles by 30%.
- Built real-time ingestion pipelines with Kafka, Pub/Sub, and Dataflow, achieving data latency of under 5 seconds for streaming inference.
- Delivered fraud prediction and churn analysis solutions used by 3 separate teams.
- Increased binary classification F1-score by 11% through Bayesian Optimization and hyperparameter tuning.
- Integrated generative AI capabilities via PaLM and Vertex LLM APIs, reducing manual ticket routing overhead by 35%.
- Implemented model drift monitoring and retraining triggers, reducing feature leakage incidents by 80%.
- Containerized models with Docker and deployed to GKE for reliable, scalable performance during usage spikes.
- Implemented CI/CD using GitLab, Jenkins, Terraform—decreasing deployment cycle from 2 days to 4 hours.
- Created Looker Studio and Tableau dashboards to visualize predictions and performance across 20+ datasets.

Outcome: Automated insights delivery, saving analysts 500+ hours/month, and improved system-wide reliability.
TOOLS: Python, Vertex AI, TensorFlow, PaLM API, BigQuery, Kafka, Cloud Storage, AutoML Tables, GitLab, Docker, Jenkins, Airflow, Looker, SQL, GKE.

Software Engineer Intern
Walmart – Remote
February 2020 – March 2021

- Developed AI-driven document classification tool using scikit-learn, TF-IDF, and Naïve Bayes, classifying 50,000+ documents with average accuracy of 88%.
- Collaborated on data cleaning and preprocessing for internal structured/unstructured data, and contributed to EDA across 7+ datasets.
- Built sentiment analysis models for product review mining with VADER and LSTM, driving 15% accuracy improvement.
- Deployed a Flask-based web app prototype, enabling non-technical teams to test ML predictions.
- Documented project progress, code, and evaluation metrics using Confluence, supporting final client review and feedback sessions.
- Participated in internal hackathons; contributed to chatbot prototype that placed in the top 3 ideas company-wide.

Outcome: Completed a full-cycle ML project from data wrangling through deployment as part of a production prototype.
TOOLS: Python, scikit-learn, NLTK, TextBlob, LSTM, Flask, TF-IDF, PyTest, GitHub, Google Colab, Seaborn, Docker, Jupyter, Rasa, VADER, Pandas, Matplotlib, Plotly.

EDUCATION

- Master of Science, Cybersecurity Operations and Warfare – Webster University, USA – 2024 – GPA: 3.67/4.0
- Bachelor of Technology, Computer Science – KL University, India – 2022 – GPA: 3.5/4.0

ADDITIONAL SKILLS

- Programming: Python, R, SQL, Bash, C
- ML/AI Frameworks: TensorFlow, PyTorch, Keras, OpenCV, Scikit-learn, Hugging Face Transformers, LangChain
- Cloud Platforms: AWS (SageMaker, Lambda, Bedrock, EC2, S3, Redshift), GCP (Vertex AI, AutoML, BigQuery, Pub/Sub), Azure ML
- Data & Storage: MongoDB, MySQL, PostgreSQL, Oracle, Google Cloud Storage, DynamoDB
- MLOps & DevOps: Docker, Kubernetes, Terraform, Jenkins, GitHub Actions, GitLab CI/CD, Bitbucket Pipelines
- Data Engineering: Apache Kafka, Apache Airflow, Google Dataflow, Spark, Pandas, NumPy
- Visualization & BI: Tableau, Power BI, Streamlit, Plotly, Grafana
- Web Frameworks: Django, Flask, FastAPI, Django Rest Framework
- Tools: PyCharm, JIRA, Confluence, Git, Toad, Putty, Prometheus
- Testing: PyTest, UnitTest
- Prompt Engineering: Tokenization, Bedrock APIs, context-aware prompt tuning, fallback handling
- Agentic AI system development
- Retrieval-augmented generation (RAG) design
- Reusable Python package development for ML automation
- Experience with Databricks for scalable analytics

RELEVANT STATISTICS

- 5+ years of AI/ML engineering experience
- 60% improvement in operational efficiency (CVS Pharmacy)
- 500+ hours/month saved via automation (Guild Mortgage)
- 99.99% uptime in production deployments